

Lab Assignment 07



Inspiring Excellence

Course Code:	CSE111
Course Title:	Programming Language II
Topic:	Array Manipulation and Inheritance Basics
Number of Tasks:	9

[Submit all the Coding Tasks (Task 1 to 8) in the Google Form shared on buX before the next lab. Submit the Tracing Tasks (Task 9) handwritten to your Lab Instructors at the beginning of the lab]

[You are not allowed to change the driver/given codes of any of the tasks]

Task 1

Design the **BrowserHistory** class so that the given output is produced.

Driver Code	Output
<pre>public class BrowserHistoryTester { public static void main(String[] args){ BrowserHistory chrome = new BrowserHistory(3); System.out.println(chrome.count + " web pages visited."); System.out.println("1====="); chrome.showHistory(); System.out.println("2====="); chrome.currentWeb(); System.out.println("3====="); chrome.visitWeb("www.google.com"); chrome.visitWeb("www.facebook.com"); chrome.visitWeb("www.youtube.com"); chrome.visitWeb("www.reddit.com"); System.out.println("4====="); System.out.println(chrome.count + " web pages visited."); System.out.println("5====="); chrome.showHistory(); System.out.println("6====="); chrome.currentWeb(); System.out.println("7====="); chrome.back(); System.out.println("8====="); System.out.println(chrome.count + " web pages visited."); System.out.println("9====="); chrome.currentWeb(); System.out.println("10====="); chrome.back(); System.out.println("11====="); chrome.currentWeb(); System.out.println("12====="); chrome.back(); System.out.println("13====="); chrome.currentWeb(); System.out.println("14====="); System.out.println(chrome.count + " web pages visited."); } }</pre>	<pre>0 web pages visited. 1===== No web pages visited yet. 2===== You have not visited any website yet! 3===== Visited: www.google.com Visited: www.facebook.com Visited: www.youtube.com History is full. Cannot visit www.reddit.com web page. 4===== 3 web pages visited. 5===== Browser History: www.google.com www.facebook.com www.youtube.com 6===== Current web page: www.youtube.com 7===== 8===== 2 web pages visited. 9===== Current web page: www.facebook.com 10===== 11===== Current web page: www.google.com 12===== 13===== You have not visited any website yet! 14===== 0 web pages visited.</pre>

Task 2

Design the **CustomerService** class so that the given output is produced.

Driver Code	Output
<pre>public class CustomerServiceTester { public static void main(String[] args) { CustomerService service = new CustomerService(3); service.serveCustomer(); System.out.println("1====="); service.addCustomer("Arthur"); service.addCustomer("Bruce"); service.addCustomer("Clark"); service.addCustomer("Kara"); System.out.println("2====="); service.serveCustomer(); service.serveCustomer(); System.out.println("3====="); service.addCustomer("Diana"); service.addCustomer("Victor"); service.addCustomer("Harley"); System.out.println("4====="); service.serveCustomer(); service.serveCustomer(); service.serveCustomer(); service.serveCustomer(); } }</pre>	<pre>No customers to serve 1===== Added: Arthur Added: Bruce Added: Clark Queue is full. Cannot add Kara 2===== Serving Arthur Serving Bruce 3===== Added: Diana Added: Victor Queue is full. Cannot add Harley 4===== Serving Clark Serving Diana Serving Victor No customers to serve</pre>

Task 3

Complete the class **Circle** so that the desired outputs are generated properly.

Given Code	Expected Output
<pre>public class shapeTester { public static void main(String[] args) { Shape s = new Shape(); s.name = "Mobius Strip"; s.color = "Blue"; s.displayInfo(); System.out.println("====="); Circle c = new Circle(); System.out.println("====="); c.name = "Circle"; c.color = "Red"; c.radius = 5; c.displayInfo(); System.out.println("====="); c.area(); } } public class Shape { public String name; public String color; public void displayInfo() { System.out.printf("Name: %s\nColor: %s\n", name, color); } } public class Circle extends Shape { //Your Code Here }</pre>	<pre>Name: Mobius Strip Color: Blue ===== Name: Circle Color: Red ===== Area of Red Circle: 78.54</pre>

Task 4

Complete the class **Dog** so that the desired outputs are generated properly.

Given Code	Expected Output
<pre>public class AnimalTester{ public static void main(String args[]){ Animal a1 = new Animal(); System.out.println("1-----"); a1.details(); System.out.println("2-----"); Dog d1 = new Dog(); d1.name = "Pammy"; System.out.println("3-----"); System.out.println("Name: " + d1.getName()); d1.details(); System.out.println("4-----"); d1.updateSound("Bark"); System.out.println("5-----"); d1.details(); } } public class Animal{ public int legs = 4; public String sound = "Not defined"; public void details(){ System.out.println("Legs: "+legs); System.out.println("Sound: "+sound); } } public class Dog extends Animal{ //Your Code Here }</pre>	<pre>1----- Legs: 4 Sound: Not defined 2----- The dog says hello! 3----- Name: Pammy Legs: 4 Sound: Not defined 4----- 5----- Legs: 4 Sound: Bark</pre>

Task 5

Given the following classes, write the code for the **Cricket_Tournament** and the **Tennis_Tournament** classes derived from **Tournament** so that the following output is generated.

Given Code	Expected Output
<pre>public class Tester5 { public static void main(String[] args) { Cricket_Tournament ct1 = new Cricket_Tournament(); System.out.println(ct1.detail()); System.out.println("-----"); Cricket_Tournament ct2 = new Cricket_Tournament("IPL", 10, "t20"); System.out.println(ct2.detail()); System.out.println("-----"); Tennis_Tournament tt = new Tennis_Tournament("Roland Garros", 128); System.out.println(tt.detail()); } } public class Tournament { private String name; public Tournament() { this.name = "Default"; } public Tournament(String name) { this.name = name; } public void setName(String name){ this.name = name; } public String getName(){ return this.name; } }</pre>	<pre>Cricket Tournament Name: Default Number of Teams: 0 Type: No type ----- Cricket Tournament Name: IPL Number of Teams: 10 Type: t20 ----- Tennis Tournament Name: Roland Garros Number of Players: 128</pre>

Task 6

Given the following classes, write the code for the **Book** and the **CD** class so that the following output is printed.

Given Code	Expected Output
<pre>public class Tester6 { public static void main(String[] args) { Book book = new Book(1, "The Alchemist", 500, "97806", "HarperCollins"); System.out.println(book.printDetail()); System.out.println("-----"); CD cd = new CD(2, "Shotto", 300, "Warfaze", 50, "Hard Rock"); System.out.println(cd.printDetail()); } } class Product { private int id; private String title; private int price; public Product(int id, String title, int price) { this.id = id; this.title = title; this.price = price; } public String getIdTitlePrice() { return "ID: " + id + " Title: " + title + " Price: " + price; } }</pre>	<pre>ID: 1 Title: The Alchemist Price: 500 ISBN: 97806 Publisher: HarperCollins ----- ID: 2 Title: Shotto Price: 300 Band: Warfaze Duration: 50 minutes Genre: Hard Rock</pre>

Task 7

Given the following classes, write the code for the **CSEStudent** class derived from **Student** so that the following output is generated.

Given Code	Expected Output
<pre>public class StudentTester{ public static void main (String args[]){ CSEStudent.details(); System.out.println("1-----"); CSEStudent s1 = new CSEStudent("Bob", 23); s1.info(); System.out.println("2-----"); CSEStudent s2 = new CSEStudent("Don", 33); s2.info(); System.out.println("3-----"); s1.addLabBasedCourse("CSE220"); s1.addLabBasedCourse("CSE221"); System.out.println("4-----"); s1.info(); System.out.println("5-----"); CSEStudent.details(); System.out.println("6-----"); s1.addLabBasedCourse("CSE230"); System.out.println("7-----"); s1.info(); System.out.println("8-----"); s2.addLabBasedCourse("CSE110"); s2.info(); } } class Student{ public String name; public int id; public String courses = ""; public Student(String n, int i){ name = n; id = i; } public void info(){ System.out.println("Name: "+name); System.out.println("ID: "+id); System.out.println("Courses: "+courses); } }</pre>	<pre>Total CSE Students: 0 Available Lab Based Courses: CSE110 CSE111 CSE220 CSE221 1----- Name: Bob ID: 23 Courses: 2----- Name: Don ID: 33 Courses: 3----- 4----- Name: Bob ID: 23 Courses: CSE220 CSE221 5----- Total CSE Students: 2 Available Lab Based Courses: CSE110 CSE111 CSE220 CSE221 6----- It is not a lab based course! 7----- Name: Bob ID: 23 Courses: CSE220 CSE221 8----- Name: Don ID: 33 Courses: CSE110</pre>

Task 8

Given the following classes, write the code for the **Player** and the **Manager** classes derived from SportsPerson class so that the following output is printed. To calculate the match earnings use the following formula:

1. Player: $(\text{total_goal} * 1000) + (\text{total_match} * 10)$
2. Manager: $\text{match_win} * 1000$

Given Code	Expected Output
<pre>public class PlayerTester { public static void main(String[] args) { Player playerOne = new Player("Al-Nassr", "Ronaldo", "Striker", 25, 32); playerOne.calculateRatio(); playerOne.printDetails(); System.out.println("-----"); Manager managerOne = new Manager("Real Madrid", "Zidane", "Manager", 25); managerOne.printDetails(); } } class SportsPerson { private String team; private String name; public String role; public double earningPerMatch; public SportsPerson(String teamName, String name, String role){ this.team = teamName; this.name = name; this.role = role; this.earningPerMatch = 0; } public String getNameTeam() { return "Name: " + name + ", Team Name: " + team; } }</pre>	<pre>Name: Ronaldo, Team Name: Al-Nassr Team Role: Striker Total Goal: 25, Total Played: 32 Goal Ratio: 0.78 Match Earning: 25320K ----- Name: Zidane, Team Name: Real Madrid Team Role: Manager Total Win: 25 Match Earning: 25000K</pre>

Task 9

1	public class A{
2	public int temp = 4;
3	public int sum = 1;
4	public int y = 2;
5	public void methodA(int m, int n){
6	int x = 0;
7	y = y + m + (temp++);
8	x = x + 2 + n;
9	sum = sum + x + y;
10	System.out.println(x + " " + y+ " " + sum);
11	}
12	}
13	public class B extends A {
14	public int x = 6;
15	public void methodB(int m, int n){
16	int y =0;
17	y = y + this.y;
18	x = this.y + 2 + temp;
19	methodA(x, y);
20	sum = x + y + super.sum;
21	System.out.println(x + " " + y+ " " + sum);
22	}
23	}

<pre> A a1 = new A(); a1.methodA(1, 1); B b1 = new B(); b1.methodB(1, 2); </pre>	x	y	sum

Ungraded Tasks (Optional)

(You don't have to submit the ungraded tasks)

Task 1

Complete the class **Student** so that the desired outputs are generated properly.

Given Code	Expected Output
<pre>public class Tester5 { public static void main(String[] args) { Student s1= new Student(1, "Matthew", 1, "Mr. XYZ"); s1.printStudentInfo(); System.out.println("====="); Student s2= new Student(2, "Wade", 1, "Mr. XYZ"); s2.printStudentInfo(); System.out.println("====="); Student s3= new Student(1, "Logan", 2, "Mr. ABC"); s3.printStudentInfo(); } } public class Section { public int sec_id; public String teacher_name; public int student_id; public String student_name; public Section(int id, String name){ this.sec_id = id; this.teacher_name = name; } public void printInfo(){ System.out.println("Section id: "+ this.sec_id); System.out.println("Teacher name: "+ this.teacher_name); } } class Student extends Section { //Your Code Here }</pre>	<pre>Student id: 1 Student name: Matthew Section id: 1 Teacher name: Mr. XYZ ===== Student id: 2 Student name: Wade Section id: 1 Teacher name: Mr. XYZ ===== Student id: 1 Student name: Logan Section id: 2 Teacher name: Mr. ABC</pre>

Task 2

Complete the class **Section** so that the desired outputs are generated properly.

Given Code	Expected Output
<pre>public class Tester6 { public static void main(String[] args) { Section sec1= new Section("Mr. XYZ"); Section sec2= new Section("Mr. ABC"); System.out.println("Number of Sections: "+ Section.section_count); System.out.println("====="); Student s1= new Student(1, "Matthew", sec1.sec_id, sec1.teacher_name); s1.printStudentInfo(); System.out.println("====="); Student s2= new Student(2, "Wade", sec1.sec_id, sec1.teacher_name); s2.printStudentInfo(); System.out.println("====="); Student s3= new Student(1, "Logan", sec2.sec_id, sec2.teacher_name); s3.printStudentInfo(); System.out.println("====="); System.out.println("Number of Students: "+ Student.student_count); } } public class Section { //Your Code Here } public class Student extends Section { public Student (int id, String name, int sec_id, String teacher_name){ student_count+=1; this.student_id = id; this.student_name = name; this.sec_id= sec_id; this.teacher_name= teacher_name; } public void printStudentInfo(){ System.out.println("Student id: "+ this.student_id); System.out.println("Student name: "+ this.student_name); System.out.println("Section id: "+ this.sec_id); System.out.println("Teacher name: "+ this.teacher_name); } }</pre>	<pre>Number of Sections: 2 ===== Student id: 1 Student name: Matthew Section id: 1 Teacher name: Mr. XYZ ===== Student id: 2 Student name: Wade Section id: 1 Teacher name: Mr. XYZ ===== Student id: 1 Student name: Logan Section id: 2 Teacher name: Mr. ABC ===== Number of Students: 3</pre>